

Visible Light Positioning With Lens Compensation for Non-Lambertian Emission

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Abstract—With greater demands for cost-effective, reliable, and highly accurate positioning, indoor wireless localisation using Visible Light Positioning (VLP) is a promising solution for future networks. One can expect VLP solutions to appear in all environments, from homes to industry; however, the existing literature primarily considers Visible Light Communication (VLC) sources with purely Lambertian emission patterns. To facilitate greater versatility within VLP solutions, this paper considers non-Lambertian sources. It evaluates practical Received Signal Strength Indicator (RSSI) data obtained during the Internet of Radio Light (IoRL) 5G Measurement Campaign conducted in a home environment using non-Lambertian Total Internal Reflection (TIR) lenses, which produce a halo lighting effect. The initial analysis explores the calibration of Lambertian source parameters against datasheet values leading to reductions in the average Positioning Error (PE) of 17% and 3% for averaged and individual RSSI measurement sets, respectively. While this highlights improvements from correct calibration, the Lambertian model proved to be unsuitable for non-Lambertian sources. In the absence of any existing non-Lambertian models, the authors proposed the Halo Lens Compensation (HLC) method to calibrate the considered non-Lambertian TIR sources correctly. The HLC further reduced PE in the calibrated results by 50% and 39%, with mean PE of 3.1 cm and 4.6 cm for averaged and individual RSSI measurement sets, respectively. In conclusion, for VLP using non-Lambertian sources, the existing Lambertian model is unsuitable. However, the proposed HLC is highly effective and achieves positioning accuracy comparable to existing literature using Lambertian sources.

Index Terms—5G Internet of Radio Light (5G IoRL), visible light positioning (VLP), non-Lambertian emission; total internal reflection (TIR) lenses.

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I. INTRODUCTION

THE EXPONENTIALLY growing numbers of IoT enabled devices and Location Based Services (LBS) offer many applications within commercial, industrial and personal fields. From highly accurate indoor navigation, Virtual Reality (VR) [1], and Augmented Reality (AR), to optimising production chains and guiding autonomous vehicles [2]. This demand reveals the necessity and importance of rapid, reliable, and highly accurate indoor wireless positioning. As published in the Third Generation Partnership Project (3GPP) releases 15 [3] and 16 [4], [5], future networks are required to achieve centimetre levels of accuracy using newly emerging processes, technologies, and spectrum. Lack of indoor coverage and requirements for higher-level accuracy dismisses the use of the well-established Global Positioning System (GPS), owing to its meter-range accuracy and inability to penetrate buildings [4], [5]. This leads to the pursuance of suitable indoor localisation methods using various available technologies such as ultrasound, infrared (IR), Bluetooth, radio-frequency identification (RFID), wireless local area network (WLAN), ultra-wideband (UWB) [8], and vision analysis [9]. Examples of such systems are Active Badge [10], CRICKET [11], and RADAR [12]. Overviews and summaries of these systems in [13], [14], [15] emphasise the limitations of cost, range, and meter-level to high-centimetre accuracy.

Unlike existing wireless positioning technologies for indoor environments, Visible Light Communications (VLC) offers vast amounts of license-free bandwidth, capable of achieving high positioning accuracy. Research into the use of VLC for indoor positioning has seen significant improvements in location performance, with accuracies in the low centimetre region. Due to its highly Line-of-Sight (LOS) nature and lack of Radio Frequency (RF) interference, VLC also lends itself to security-sensitive and medical applications. Notably, the minimal adaptation required for existing lighting infrastructure in optimal LOS coverage locations and the use of standard, low-cost, and energy-efficient LEDs makes VLC a viable solution for widespread implementation.

The Internet of Radio Light (IoRL) 5G Project [16], [17] presents a novel indoor 5G solution to achieve 10 GBps wireless data rates with under 1-millisecond latency through the hybrid use of Millimetre Waves (MMW) and VLC. This paper focuses on the Visible Light Positioning (VLP) component.

A. Existing Literature

Conventionally there are four fundamental methods of VLP: proximity, triangulation, fingerprint analysis, and image positioning. The proximity [18] method uses a grid format where each grid cell contains and is associated with a singular VLC source. In this respect, Received Signal Strength (RSS) determines the generalised target location from the most robust signal received. Albeit a straightforward solution, cell dimensions limit the resolution. Fingerprint analysis [19] estimates the target location by comparing real-time signal characteristics, referred to as fingerprints, against a pre-existing database of measured signal identifiers. Fingerprints may include but are not limited to the signal's Time of Arrival (TOA), Time Difference of Arrival (TDOA) or RSS. A variety of techniques have been introduced for appropriately associating real-time measurements. While online positioning is relatively fast, formulating a fingerprint database is often cumbersome. Furthermore, results are sensitive to environmental changes and the resolution of the database. Researchers in [19] adopt a Weighted K-Nearest Neighbours (WkNN) approach, achieving 2D accuracies within 8-0.3cm dependent on the area covered. Experiments in [29], utilising a Modified Back Propagation neural network, achieve average accuracies under 2cm from databases containing only 20 points. Triangulation uses the geometric properties of triangles for position estimation. The two subsets of triangulation, namely lateration and angulation, achieve localisation using range or angular measurements, respectively. Trilateration [20], [28], or 'multilateration' for instances with more than three anchor nodes, employs TOA, TDOA or RSS to ascertain the distances from known anchor points and the target. Due to the practical nature of VLC, anchor points are conventionally VLC source transmitters, while the target is a Photodiode (PD) receiver connected to a mobile device. Similarly, triangulation employs the signal's Angle of Arrival (AOA) [30], [31] to acquire the relative angles between the transmitters and the PD. Unlike the techniques above, Image positioning [32], [33] employs an image sensor or 'camera' instead of a regular PD. Position estimates are derived by the geometric relations between the VLC sources in 3D space and the 2D projections obtained by the image sensor. While achieving high accuracy, the image processing speed limits the sampling rate, and accuracy is sensitive to lighting conditions and pixel resolution. Experiments conducted in [32], [33] deliver sub 8cm level accuracies for image based 3D localisation.

This study employs trilateration due to its popularity, high positioning accuracy, low complexity and use of a PD in obtaining the available dataset. Additionally, due to the stringent synchronisation requirements required of time-based measurements such as TOA and TDOA, RSS-Indicator (RSSI) measurements are considered throughout, where RSSI is a relative measure of RSS with arbitrary units [34].

Within this scope, the VLP experiments conducted in [22] present promising RSS Trilateration results with accuracies below 2 cm. Researchers in [20] present a comparative simulation study of distinct and uniform lighting configurations regarding non-LOS (NLOS) impact. The study further explores the effects under variable receiver height, semi radiation angle,

and Field of View (FoV). Notably, the study also presents significant degradation in accuracy near edges and corners of the walls, which is experimentally supported by the findings in [21]. To overcome the effects of intercell interference, the authors of [25] consider multiple PD receivers for a single transmitter on a rotating platform. Generally, the RSS VLC trilateration approach considers the VLC LOS channel model, introduced later in Section II-B, to infer the relative distances from the RSS. Due to the impact of radiant and incidence angles, RSS VLC trilateration is often restricted to two dimensions (2D), where the target's height is assumed or known [20], [21], [22], [25]. Such systems are applicable for robot navigation. As presented in [26], iterative solutions at various altitudes may overcome the 2D limitation. However, this introduces additional computation and accuracy is affected by the resolution of height intervals. Alternatively, implementing additional sensors such as Inertial Measurement Units (IMU) to provide LOS path angles [27], [28] can enable 3D positioning.

Notably, regardless of the dimensions considered, VLP experiments conducted with trilateration methods are often carried out in ideal lab settings, where commercial off the shelf light sources exhibit or are assumed to exhibit conventional Lambertian emission patterns. While Lambertian model assumptions are circumvented in data-centric fingerprinting solutions [35], [36], [37], for lightweight and simple VLP use cases to develop under various scenarios and environments, non-Lambertian sources in non-ideal systems must be considered using trilateration. Research in [22], [23] and [27] experimentally model the VLC source patterns to more accurately calibrate channel parameters according to the Lambertian model. This is emphasised in [23], where experiments highlight the disparity in source parameters between identical light sources. Conversely to the widely adopted Lambertian model, it has been proposed in [24] and applied in [25] that an exponential model better suits the emission pattern due to the refractive index of LED lenses. Table I summarises existing RSS Trilateration VLP literature and the applied models. Real-world results explored within this paper highlight the adverse effects of lens structure on emission patterns for VLP trilateration solutions and the importance of correct model calibration. A solution is proposed that corrects these lens effects, specifically for lenses that cause a halo ring phenomenon. Corrections presented lead to a substantial reduction in Positioning Error (PE) and a mean PE under 5cm. The key contributions of this paper are:

- 1) The presentation and analysis of IoRL 5G 2D VLP measurement campaign results within a home environment achieving an average PE under 5cm.
- 2) The development of a calibration method to resolve VLP errors for non-Lambertian sources, specifically Halo ring lens distortions from Total Internal Reflection (TIR) semi-frosted lenses.

The rest of this paper is structured as follows. In Section II, the system overview and positioning procedure are introduced. The experimental setup and key components used are outlined in Section III. In Section IV, the process implemented for the calibration of the optimal Lambertian parameters is delivered. Initial analysis and the Lambertian model errors are discussed in

TABLE I
SUMMARY OF EXISTING VLP SOLUTIONS AND RESULTS

Dimension	Mean Positioning Error /m	Method	Tx arrangement	Rx orientation	Lambertian model	Area / ($L * W * H$) /m	Experiment/ Simulation	Ref
2D	0.0961 0.1498	RSS Multilateration	Uniform arrangement Distinct arrangement		Assumed	5*5*4	Simulation	[20]
2D	0.1800	RSS Multilateration	Uniform square	Rx horizontal	Assumed	2*1*2.4	Experiment	[21]
2D	0.0168	RSS Trilateration	Right-angled triangle	Rx horizontal	Fitted	0.2*0.2*0.15	Experiment	[22]
2D	0.0280	RSS Multilateration	Uniform square	Rx horizontal	Fitted	4.8*5.7*2.5	Experiment	[23]
2D	0.0040 0.0240	RSS Trilateration	Equilateral triangle	Rx horizontal	Exponential model	0.6*0.6*0.6	Simulation Experiment	[24]
2D	0.0065	RSS Trilateration	Single Tx	Multiple Rx horizontal	Exponential model	2*2*2	Simulation	[25]
3D	0.0730 0.0480	Iterative RSS Multilateration	4 in Uniform square 5 in Arbitrary positions	Rx horizontal	Assumed	5*5*5	Simulation	[26]
3D	90% <0.45 90% <0.7 90% <0.8 0.4000	RSS Multilateration + IMU	5 in a star configuration 5 in 2 parallel lines 5 in a single line 16 uniformly distributed	Rx horizontal with minor angular changes	Fitted	5*8*N/A 2*12*N/A 3.5*6.5*N/A	Experiment	[27]
3D	<0.2500	RSS Trilateration + Accelerometer	Triangle	variable	Assumed	5*3*3	Simulation Experiment	[28]

TABLE II
MATHEMATICAL NOTATION

Denotation	Meaning
$\ A - B\ $	The Euclidian norm between A and B
$ A $	The magnitude of A
$A^{x,y}$	A is associated with the Cartesian coordinates of x and y
A^T	The Transpose of A
A^{-1}	The inverse matrix of A (or Moore-Penrose)
$\ln(A)$	The natural logarithm of A
$A(:, B)$	All row data contained within column B of A
$A = [B:C:D]$	A is assigned values from B to D in increments of C

Section V before the Halo Lens Compensation (HLC) method is presented in Section VI and analysed in Section VII.

II. SYSTEM MODEL AND VLP MULTILATERATION PROCESS

A. System Overview

The derivation of the user position in Cartesian coordinates from RSSI measurements is considered in two stages. Firstly, the range determination stage establishes range estimates between each VLC source and the target PD. This requires the VLC channel model to define a relationship between measured RSSI and the distances. Next, given the set of nonlinear range estimates and known anchor coordinates, the Multilateration method determines a position estimate through the geometric properties of triangles. Consider the scenario shown in Fig. 1, where four transmitters, located at priori known locations, are emitting directly downwards. The i 'th transmitter is denoted by $X_i = [x_i, y_i, z_i]^T$ for $i = (A, B, C, D)$ where x_i, y_i and z_i denote the Cartesian coordinates of the centre of the i 'th transmitters LED light source. The target to be located is a receiver (Rx) module with a PD facing directly upward. The PD is denoted by $X_{PD} = [x, y, z]^T$ where x, y and z denote its Cartesian coordinates. The vertical, horizontal, and Euclidean distance between X_{PD} and X_i are given by h_i, r_i and d_i respectively.

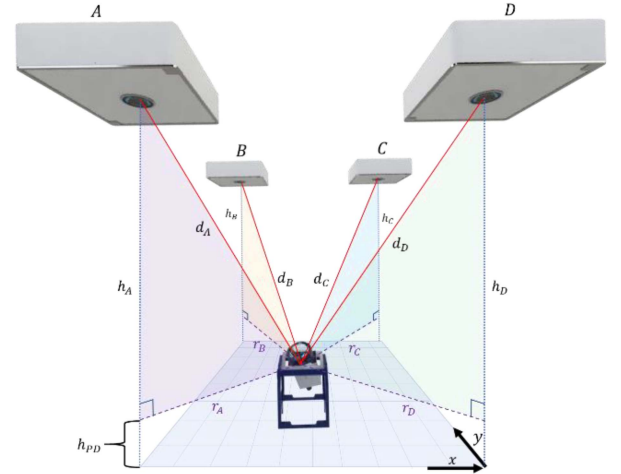


Fig. 1. Geometry of transmitter-receiver relationships used in conventional VLC Multilateration.

B. 2D VLC RSSI Channel Model and Range Estimation

Standard VLC positioning algorithms assume that the radiation pattern of a light source adheres to Lambert's cosine law, which generally states that, for any emitter, the source's intensity is proportional to the cosine of the angle between the path of radiation and the normal [38]. Using this relationship and considering only the Line of Sight (LOS) path, the RSSI and Euclidean distance (d_i) between the PD and the i 'th transmitter is described by the VLC channel model [21]:

$$P_i^R = H(0) * P_i^T, \quad (1)$$

where P_i^T and P_i^R represent the transmitted power and RSSI respectively and $H(0)$ represents the VLC LOS channel gain between the i 'th transmitter and the receiver, given as [39]:

$$H_i(0) = \frac{(m_i + 1)A_r \cos^{m_i}(\psi_i) \cos(\theta_i) T(\theta_i) G(\theta_i)}{2\pi d_i^2}, \quad (2)$$

where A_r is the priori known effective area of the PD, ψ_i and θ_i denote the angle of emission and incidence respectively, T

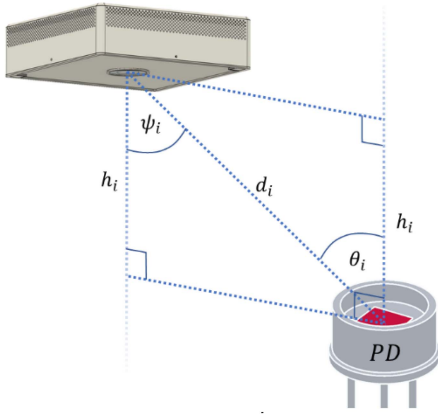


Fig. 2. Geometric relationship between the i 'th transmitter and receiver PD.

is the optical filter gain, G is the optical concentrator gain and m_i is the order of Lambertian emission given by [38]:

$$m_i = -\frac{\ln(2)}{\ln(\cos(\psi_{1/2}^i))}, \quad (3)$$

where $\psi_{1/2}^i$ is the semi-angle at half power of the source. The relationship between ψ_i and θ_i to the normal of the i 'th transmitter and receiver PD are depicted in Fig. 2. Exploiting the trigonometric relationships between the right-angled triangles of Fig. 2, one can obtain the following expressions:

$$\cos(\theta) = \cos(\psi) = \frac{h_i}{d_i}, \quad (4)$$

$$\therefore \cos^{m_i}(\psi) = \left(\frac{h_i}{d_i}\right)^{m_i} = \frac{h_i^{m_i}}{d_i^{m_i}}, \quad (5)$$

Substituting (4) and (5) into (2) and then (2) into (1) and re-arranging for d_i produces the well-established VLP range equation (6) [39].

$$d_i = \|X_i - X_{PD}\| = \sqrt[2]{\left(\frac{(m_i+1)A_r h_i^{(m_i+1)} T(\theta_i) G(\theta_i)}{2\pi}\right) \frac{P_i^T}{P_i^R}} \quad (6)$$

Equation (6) eliminates the need for known emission and incidence angles but introduces a known vertical height assumption.

C. Multilateration Position Estimation

Given the known vertical height difference h_i and estimated Euclidian distance d_i , the Pythagoras theorem can be applied to determine the horizontal distance component r_i (7) [21]. Reducing the overall problem to a 2D multilateration solution.

$$r_i = \sqrt{d_i^2 - h_i^2}, \quad (7)$$

where,

$$r_i = \sqrt{(x - x_i)^2 + (y - y_i)^2}, \quad (8)$$

From (8) squaring both sides, expanding and re-arrangement leads to:

$$(x^2 + y^2) - 2x_i x - 2y_i y = r_i^2 - x_i^2 - y_i^2, \quad (9)$$

Repeating (6)–(9) for each transmitter provides an overdetermined set of equations expressed in matrix form [21]:

$$A\hat{\vartheta} = \hat{b}, \quad (10)$$

where,

$$A = \begin{bmatrix} -2x_1 & -2y_1 & 1 \\ \vdots & \vdots & \vdots \\ -2x_4 & -2y_4 & 1 \end{bmatrix}, \hat{\vartheta} = \begin{bmatrix} x \\ y \\ x^2 + y^2 \end{bmatrix}, \quad (11)$$

$$\hat{b} = \begin{bmatrix} r_1^2 - x_1^2 - y_1^2 \\ \vdots \\ r_4^2 - x_4^2 - y_4^2 \end{bmatrix},$$

the 2D coordinates of the PD ($\hat{X}_{PD}$) can be solved for using the Least Squares (LS) solution to determine $\hat{\vartheta}$. The LS solution is given by [21], [40]:

$$\hat{\vartheta} = (A^T A)^{-1} A^T \hat{b}, \quad (12)$$

III. IORL VLC MEASUREMENT CAMPAIGN

The complete description of the IoRL Measurement Campaign is detailed in [41]; however, for clarity and practical significance, what follows is a summation of the experimental methodology and key hardware used to acquire the VLC RSSI measurement datasets evaluated in this paper.

A. Experimental Setup

The IoRL measurement campaign was conducted at the Building Research Establishment (BRE) in a furnished open plan sitting room of a home environment. The Experimental setup corresponds to that of the scenario described in Section II. To simplify logistics, the hardware was installed on a 0.4m wide square cross-section aluminium extrusion frame having an overall dimension of (2.5 * 2.5 * 2), in the form ($L * W * H$)/m. This experiment was conducted on a horizontal centimetre grid on a flat surface 0.7 m above the ground. The origin point was roughly in the centre of the grid. The four VLC transmitters - labelled A, B, C and D - are situated above the grid with the respective x, y, z /m Cartesian coordinates $(-29, -0.26, 1.424)$, $(-0.285, 0.275, 1.524)$, $(0.235, 0.275, 1.524)$ and $(0.24, -0.28, 1.424)$, where z describes the height from the grid surface. The experimental setup is illustrated in Fig. 3. The VLC link uses Orthogonal Frequency Division Multiplexing (OFDM) with a carrier frequency of 15 MHz, 10 MHz bandwidth and 30 kHz subcarrier spacing. Due to the IoRL project originally being devised to be compliant to 3GPP release 15 (LTE) standard, Hermitian symmetry is imposed on the OFDM subcarriers with a DC bias added to the bipolar OFDM time-domain signal, rendering the signal real and non-negative as required for OFDM symbols transmission over VLC medium.

B. VLC Source Lenses

A circular chip on board (COB) light source is used in conjunction with a 70 mm diameter TIR lens [42], as shown in Fig. 4. As in Fig. 4a, the optics incorporate a Fresnel lens

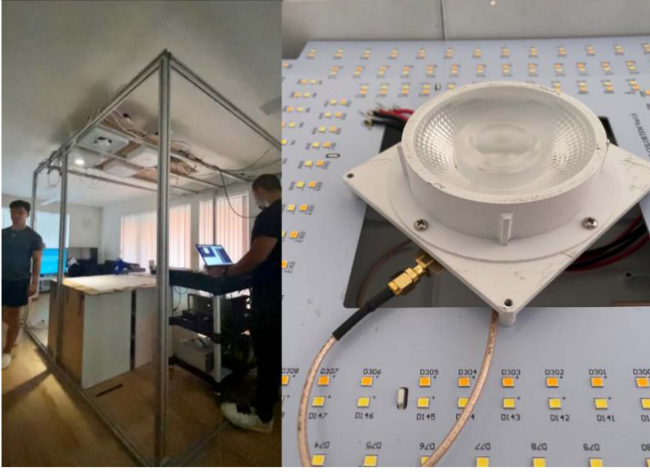


Fig. 3. IoRL Measurement Campaign Experimental Setup. (Left) four light sources emitting downward onto a raised grid, only one VLC source is active at a time. (Right) central VLC source and lens used surrounded by non-active LEDs.

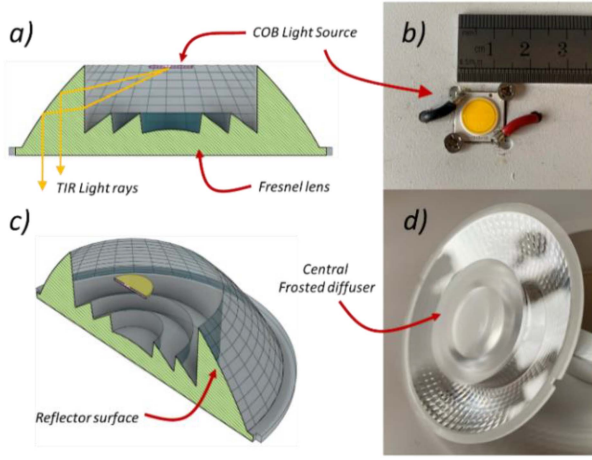


Fig. 4. TIR lens and COB source. a) Cross-section view; b) COB VLC source; c) Angled cross-section view; d) Flat face of the lens with central frosted diffuser.

to reduce the overall form factor and evenly distribute near incident light. The COB source is surrounded by a reflective surface that captures wider rays of light and, through total internal reflection, directs them in a collimated and controlled manner. Using this TIR lens, the emission pattern becomes narrower and more direct. The frosted area in Fig. 4d acts as a diffuser for light emitted through the Fresnel lens region, thus reducing light intensity at low radiation angles.

The datasheet for the combined light source and lens includes the Luminous Intensity Graph (LIG) of Fig. 5 for two ways of orienting the light. Different orientations produce half-powers of 22.5° and 21.9°, which are averaged out to 22.2°

C. Receiver Gimbal

The VLC signal is focused onto receiver PD using a large convex glass lens. This lens requires a focal point separation of 1.5cm from the PD. For angles of incidence greater than

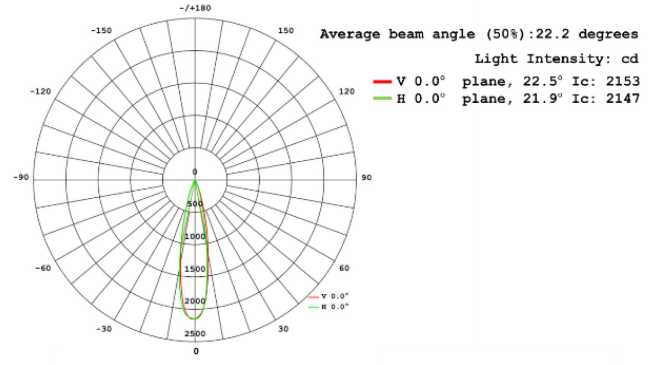


Fig. 5. Luminous Intensity Graph (LIG) of light source and lens from the datasheet.

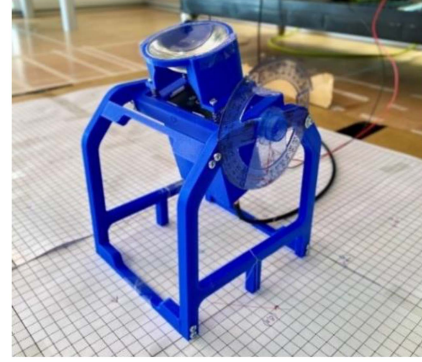


Fig. 6. IoRL VLC receiver gimbal with large convex lens for focusing VLC signal, fixed protractors for orientation adjustment and single thread crosshair for accurate positioning.

half a degree,¹ the lens's extensive focal length prevents the VLC signal from reaching the PD. Consequently, the receiver module is oriented directly toward one active source at a time. To achieve stable measurements, the receiver must be reoriented to face each transmitter without interfering with PD positioning. Thus, the receiver module is mounted in a gimbal, as depicted in Fig. 6. Notably, the elimination of the angle of incidence will result in more accurate positioning due to improved LOS RSSI quality and one must account for the gimbal height ($h_{PD} = 15.5$ cm) in the following solutions. Moreover, calculations in Section II-B must be modified to account for an incidence angle consistently equal to zero ($\theta = 0$). Firstly, since θ is constant, the optical concentrator gain $G(\theta)$ is constant for all measurements and is omitted due to lack of influence in the overall localisation process. Secondly, in formula (4) the relationship becomes $\cos(\theta) = 1$, and the subsequent calculation of d_i using (1), (2), (4) and (5) results in:

$$d_i = \|X_i - X_{PD}\| = \sqrt{\left(\frac{(m_i+1)A_r h_i^{(m_i)}}{2\pi} \right) \frac{P_{T_i}}{P_{R_i}}}, \quad (13)$$

Due to the lack of any applied optical filter at the receiver, the optical filter gain $T(\theta) = 1$ and is omitted from (13).

¹Alternative smaller and wide-angle lenses significantly reduced VLC signal reception.

TABLE III
EXAMPLE OF A SINGLE DF OF RECORDED RSSI MEASUREMENTS FOR
THE GLOBAL COORDINATES x, y

DataFile – Rx position $X_{PD} = (x, y, z = 0)$			
Timestamp	sub_1	...	sub_{192}
$t_1^{x,y}$	RSSI($t_1^{x,y}, sub_1$)	...	RSSI($t_1^{x,y}, sub_{192}$)
$\vdots$	$\vdots$	...	$\vdots$
$t_{40}^{x,y}$	RSSI($t_{40}^{x,y}, sub_1$)	...	RSSI($t_{40}^{x,y}, sub_{192}$)

TABLE IV
SUMMARY OF EXPERIMENTAL VARIABLES

Parameter	Value	
Effective Area of PD (A_r)/ m^2	$7.07e^{-4}$	
PD height (h_{PD})/ m	0.155	
Grid Height(h_g)/ m	0.7	
VLC Source $\psi_{1/2}^\circ$	22.2	
VLC source coordinates (x, y, z)/ m	X_A	(-0.29, -0.26, 2.124)
	X_B	(-0.285, 0.275, 2.134)
	X_C	(0.235, 0.275, 2.124)
	X_D	(0.24, -0.28, 2.124)
Vertical difference between PD and Lights (h_L)/ m	h_A & h_C	1.2690
	h_B & h_D	1.2790
Grid x-axis / m	[-0.3: 0.1: 0.4]	
Grid y-axis / m	[-0.4: 0.1: 0.3]	
Positioning area ($L * W$)/ m	$0.8 * 0.8$	
Number of coordinates measured (N_i)	A	64
	B	43
	C	63
	D	59

D. Methodology

Each transmitter's VLC source is mapped using a plumb line on the 2D grid; these points represent each light's origin. Reference RSSI measurements are taken at the marked origin points directly beneath each light source. A custom Python program is used to store RSSI measurements on the User Equipment (UE) connected to the VLC Rx. In total, the receiver is positioned 64 times across both the x and y axes, at intervals of 10 cm. At each position, the receiver gimbal is orientated towards the currently active VLC transmitter. In order to ensure LOS to the PD, mounted protractors are used, and visual inspection is performed. Prior to taking continuous RSSI measurements, the equipment is left stationary to allow the signal to stabilize. While the gimbal is fixed in place, the next VLC source is activated, and the first is disconnected. After the receiver gimbal has been reoriented, the process is repeated until all four light sources have been measured for RSSI. Finally, a collection of (x, y) coordinate dependent RSSI Data Files ($DF_i^{x,y}$) is obtained for each VLC source ($i = A, B, C, D$). Moreover, a RSSI reference Data File (DF_i^{REF}) is obtained at each source's origin point. Table III illustrates a sample of a single $DF_i^{x,y}$ where for each instant of time (t) the RSSI of 192 subcarriers (sub) is given. Each $DF_i^{x,y}$ contains 40 measured time points. For subcarrier j the frequency, in Hz, is given by $f_j = 9.24E6 + j(6E4)$.

Table IV summarises the key experimental variables.

Remark 1: N_i varies between sources as some sources failed to produce a stable signal at certain coordinates.

IV. CALIBRATION: OPTIMAL VARIABLE IDENTIFICATION

The performance of a VLP system is mainly dependent on the $\psi_{1/2}$ of the VLC source as this influences the relationship between the RSSI and estimated Euclidian distance; this is evident within (13) and has been remarked in the literature [23]. A source's datasheet typically provides a $\psi_{1/2}$ value, as in Section III-B. Nevertheless, calibrating for $\psi_{1/2}$ is discussed in [23] and demonstrates that identical lights may have different values. Therefore, in the following sections, $\psi_{1/2}^i$ values are empirically determined. Additionally, literature typically advocates using the highest power subcarriers [21], [22] but the measurement campaign captured RSSI data for 192 subcarriers; as a result, subcarrier options are discussed briefly. This section describes the calibration phase to determine the optimal Calibration Parameters $CP_i = [\psi_{1/2}^i, sub_i]$ for each source. This calibration method is closely related to the work presented in [23]. Except for the DF_i^{REF} , all $DF_i^{x,y}$ s are halved, resulting in Calibration Data Files $CDF_i^{x,y}$ and Measurement Data Files $MDF_i^{x,y}$ containing 20 sets (rows) of measurements each.

A. Calibration Process

Here an exhaustive search approach is used to determine CP_i based on numerous sub_i and $\psi_{1/2}^i$ combinations. Where $\psi_{1/2}^i = [5:0.2:60]$ is given in degrees and $sub_i = [1:1:192]$ refers to the column in Table III.

1) *Compute P_i^T :* The transmit power of the i 'th source is computed using the DF_i^{REF} . Since the DF_i^{REF} is obtained directly below the i 'th source, ψ and θ equate to zero and (2) reduces to (14).

$$H_i(0) = \frac{(m_i+1)A_r}{2\pi d_i^2}, \quad (14)$$

In which m_i is computed from (3) and the currently assigned $\psi_{1/2}$ value. In addition, d_i is known and equal to h_i at the reference position. P_i^T is subsequently calculated by rearranging (1) and substituting (14) for the channel gain. The RSSI at the reference point (P_i^{REF}) is taken to be the average RSSI of the assigned sub_i column within the DF_i^{REF} .

2) *Obtain the Optimal Parameter Values:* The optimum CP_i is the set of values that minimizes the average range error in the calibration dataset. To determine the optimal CP_i , range estimation is performed using the obtained P_i^T value from Section IV-A1) and the range calculation given by (13) for all measured coordinates. During this calibration phase, RSSI for each position is given by the respective $CDF_i^{x,y}$ and the mean of the assigned sub_i column $CDF_i^{x,y}(:, sub_i)$. The absolute difference between the resultant estimated range ($\hat{d}_i^{x,y}$) and the true range $d_i^{x,y}$ provides the position range error.

$$\varepsilon_{Range,i}^{x,y} = \left| \hat{d}_i^{x,y} - d_i^{x,y} \right|, \quad (15)$$

The position range error for all measured coordinates are averaged, as in (16), to determine the CP_i values that yield

Algorithm 1: Calibration: Exhaustive Search for Optimal CP_i

```

For  $\psi_{1/2}^i = [5:0.2:60]$ 
  For  $sub_i = [1:1:192]$ 
    Determine  $P_i^T$ 
    Calculate  $m_i$  using  $\psi_{1/2}^i$  in (3).
    Determine  $P_i^{REF}$  from  $DF_i^{REF}$ 's using the
    averaged  $sub_i$  column data.
    Calculate  $P_i^T$  from (14) and (1).
    Determine mean error
    For each  $(x, y)$  in the set of  $N_i$  retrieved
    measurements:
      Attain  $P_i^{R(x,y)}$  from the average RSSI of
       $CDF_i^{x,y}(:, sub_i)$ .
      Use (13) to estimate the distance  $\hat{d}_i^{x,y}$ .
      Determine range error  $\varepsilon_{Range,i}^{x,y}$  from (15).
    End
    Determine the mean range error (16).
  End
End
Ascertain the  $CP_i$  that yields the lowest average range
error

```

the minimum mean range error ($\varepsilon_{Range}^{\psi_{1/2}, Sub}$).

$$\varepsilon_{Range,i}^{\psi_{1/2}, sub} = \frac{\left(\sum_{j=1}^{N_i} \varepsilon_{Range,i}^{x,y} \right)}{N_i}, \quad (16)$$

This process is summarised in Algorithm 1 and is repeated for each light.

B. Position Estimation

Collecting the set of optimal parameters $CP = [CP_A; CP_B; CP_C; CP_D]$, position estimation is carried out using the multilateration process described in Section II-C where the P_i^R in (13) is determined using the available measurement dataset in two ways. The first method simply takes RSSI to be the average of the sub_i column within the respective $MDF_i^{x,y}$. Alternatively, the second approach considers the use of individual RSSI measurements. However, as detailed in Section III-D, the acquisition of these measurements is asynchronous and therefore the combination of individual RSSI measurements for multilateration is arbitrary. In this respect, individual RSSI values from all VLC sources are combined in an exhaustive approach.

$$P_i^{R(x,y)} = \begin{cases} \text{Avg}\left(MDF_i^{x,y}(:, sub_i)\right), & \text{Averaged} \\ MDF_i^{x,y}([1:1:20], sub_i), & \text{Individual} \end{cases} \quad (17)$$

The PE is given by the Euclidian distance between the estimated PD position ($\hat{X}_{PD}$) and the true position (X_{PD}).

$$\varepsilon_{Position}^{x,y} = \left\| \hat{X}_{PD}^{x,y} - X_{PD}^{x,y} \right\|, \quad (18)$$

The position estimation process is summarised in Algorithm 2.

Algorithm 2: Position Estimation

```

For  $i = A, B, C, D$ :
  Determine  $P_i^T$ 
  Calculate  $m_i$  using  $\psi_{1/2}^i$  in (3).
  Determine  $P_i^{REF}$  using averaged  $sub_i$  column from
   $DF_i^{REF}$ .
  Calculate  $P_i^T$  using (14) and (1).
End
Position Estimate Error (Averaged)
For each  $(x, y)$  in the set of  $N_i$  retrieved measurements:
  For  $i = A, B, C, D$ :
    Obtain  $P_i^{R(x,y)}$  from  $mean(MDF_i^{x,y}(:, sub_i))$ .
    Estimate the distance  $\hat{d}_i^{x,y}$  using (13).
  End
  Use (7)–(12) to estimate the position coordinates
   $\hat{X}_{PD}^{x,y}$ .
  Calculate the PE using (18).
End
Position Estimate Error (Individual)
For each  $(x, y)$  in the set of retrieved measurements:
  For  $P_A^R = MDF_A^{x,y}([1:1:20], sub_i)$ :
    For  $P_B^R = MDF_B^{x,y}([1:1:20], sub_i)$ :
      For  $P_C^R = MDF_C^{x,y}([1:1:20], sub_i)$ :
        For  $P_D^R = MDF_D^{x,y}([1:1:20], sub_i)$ :
          Estimate  $\hat{d}_i^{x,y}$  for  $i = A, B, C, D$  using
          (13).
          Calculate  $\hat{X}_{PD}^{x,y}$  using (7)–(12).
        End
      End
    End
  End
  End
  Average the  $\hat{X}_{PD}^{x,y}$  estimates.
  Calculate the PE using (18).
End

```

Remark 2: Following from Remark 1, not all positions obtain range estimates from all four sources, in which case trilateration is performed. In the absence of three range estimates, no estimate is given.

V. INITIAL ANALYSIS OF RESULTS

The following analysis of calibrated results, obtained from using CP values, is compared against the given datasheet $\psi_{1/2}$ value (22.2°), hereafter referred to as the *Control*. Optimal sub_i values are calculated for the *Control* through a fixed $\psi_{1/2}$ iteration of Algorithm 1.

A. Calibration of CP_i

The optimal CP_i values obtained from Algorithm 1 are presented in Table V and the output of Algorithm 1 is shown in Fig. 7, where the response is highly sensitive to changes in $\psi_{1/2}$ values and an optimal value is clearly discernible. In contrast, the subcarrier selection appears to have a less significant effect on range error. The standard deviation within the average range errors across all possible subcarriers at their

TABLE V
OPTIMAL CP_i

A		B		C		D	
$\psi_{1/2}^A / ^\circ$	Sub _A	$\psi_{1/2}^B / ^\circ$	Sub _B	$\psi_{1/2}^C / ^\circ$	Sub _C	$\psi_{1/2}^D / ^\circ$	Sub _C
25.6	142	24.4	165	24	110	25.8	87
Control values							
22.2	135	22.2	100	22.2	53	22.2	61

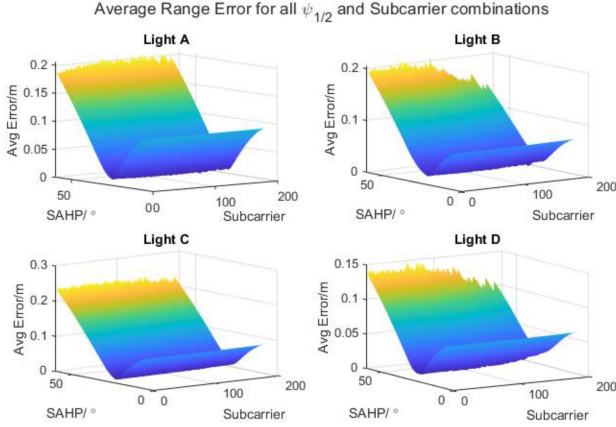


Fig. 7. Average range errors across all measured points for all subcarriers and Semi-Angle at Half Powers (SAHP) $\psi_{1/2}$.

TABLE VI
LIGHT SOURCE RANGE ERROR RESULTS

A		B		C		D	
Mean /cm	Std. /cm	Mean /cm	Std. /cm	Mean /cm	Std. /cm	Mean /cm	Std. /cm
Control Results							
2.37	1.75	1.34	1.63	1.27	1.25	1.44	1.67
Calibrated Results							
0.99	1.35	1.33	1.82	1.03	1.36	1.28	1.77

respective optimal $\psi_{1/2}^i$ values is 1.1 mm, 2.0 mm, 0.8 mm and 4.1 mm for Lights A, B, C and D, respectively.

B. Range Estimation Errors

To compare individual lights to one another, the mean range errors using the average RSSI from each $MDF_i^{x,y}(:, sub_i)$ are given in Table VI. From Table VI, it is evident that not all lights perform equally as Light A and C obtain the lowest mean error and standard deviation across all points, followed by Light D and Light B. This pattern correlates to the coverage measured during the measurement campaign in [41]. The calibration process has proven to reduce the mean range errors for lights A, B, C and D by 58%, 1%, 19% and 11%, respectively.²

C. Identifying Irregularities in the Lambertian Model

Fig. 8 presents the distribution of non-absolute range errors for each light across all measured points. Noticeably, each light exhibits a common arc region in which the range errors are negative, as indicated by the darker areas. A negative range

²For brevity, tabulated data is given to two decimal places, however percentage calculations are computed using six decimal places.

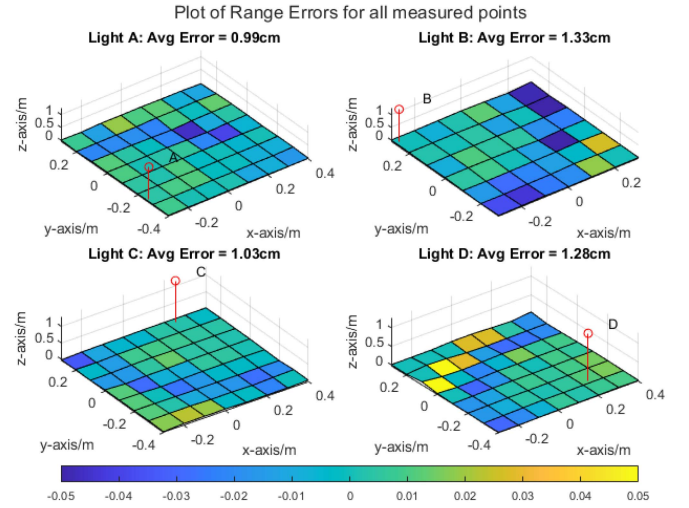


Fig. 8. Individual light ranging errors across all points.

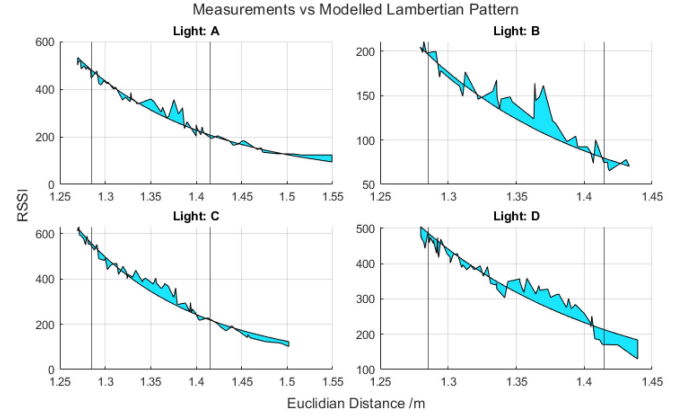


Fig. 9. Measured RSSI against distance compared against the fitted Lambertian model. Coloured regions indicate the disparity between the model and measurements. Black lines indicate exaggerated assumptions of the HR.

error from the non-absolute adaptation of (14) indicates the estimated distance is less than the true distance. Given the negative correlation between P_i^R and $\hat{d}_i$ in (13), a reduced distance indicates an increased RSSI. This region indicates an irregularity in the Lambertian relationship between P_i^R and $\hat{d}_i$ given by (13). For clarity, Fig. 9 depicts the differences between the RSSI measurements within the $MDF_i^{x,y}$ s, at given Euclidian distances, and the expected RSSI, according to the Lambertian models' CP_i values. From Fig. 9, the area at which the RSSI are higher than the expected RSSI is more evident and occurs at approximately the same region of Euclidian distances for each light. However, the magnitude of the disparity between the modelled and measured RSSI varies among the individual lights, justifying the difference in Range error percentage improvements.

Since this region is consistent throughout all lights and at seemingly constant radii, it is reasonable to conclude that the area exists due to the structure of the source's lens. The lens focuses light in a non-Lambertian manner causing a region of higher-than-expected light intensity and higher RSSI measurements. Observing the lens characteristics presented in

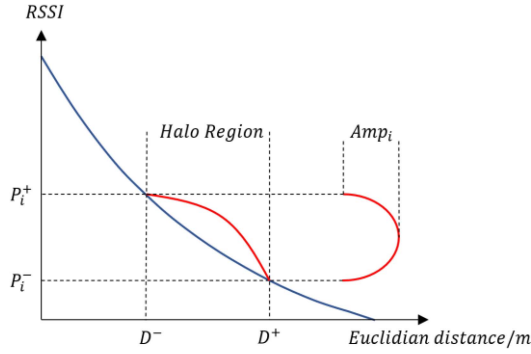


Fig. 10. A general overview of the proposed solution where a peak is included across the RSSI range that relates to the HR to compensate for the higher RSSI values.

Section III-B and Fig. 4; the cause of this non-Lambertian emission pattern is likely to be both the central diffuser, which reduces the intensity of incident light rays, and the lens reflector that refocuses light directly downward as opposed to outward as Lambert's cosine function suggests. The region of higher RSSI is hereafter referred to as the 'Halo Region' (HR). This draws further attention to the need for calibrating when performing VLP processes as not all light emission patterns are Lambertian. The HR can be seen to cause errors in two significant ways. Firstly, the Lambertian assumption is inappropriate for this distribution and will consistently misrepresent the distance. Secondly, during the calibration phase, described in Section IV, the optimal CP_i values reflect the fit that reduces the overall error. This leads to a poor fit for both the ranges within the HR and those outside it, as the Lambertian model is forced to accommodate all values.

VI. PROPOSED LENS HALO COMPENSATION

To reduce the effects of the HR, the model based on the Lambertian assumption should be corrected to reflect the genuine relationship between RSSI and distance. The following section describes the proposed calibration procedure, termed the Halo Lens Compensation (HLC) method.

A. Overview of the Proposed Solution

The proposed HLC method fits an ideal Lambertian distribution across all points outside the HR and adds a piecewise skewed peak across the HR, with a calibrated amplitude and skew for tuning. Fig. 10 illustrates the solution concept to adjust for the higher RSSI values within the HR using a fitted peak.

The HR is defined by the minimum and maximum Euclidian distances at which it occurs, represented by D^- and D^+ respectively. P_i^+ and P_i^- denote the VLC source-specific region limits of RSSI values that correlate to D^- and D^+ respectively. For notational convenience $P_i^r = [P_i^+, P_i^-]$, $D = [D^-, D^+]$ and the Updated Calibrated Parameters (UCP_i) = $[\psi_{1/2}^i, Sub_i, P_i^r, Amp_i, S_i]^T$. The peak to be applied is generated using (19) where Amp_i is a scaling factor relating to the amplitude of the peak and S_i denotes the Skew factor as illustrated

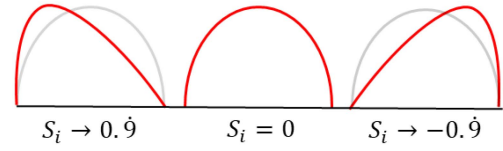


Fig. 11. Effects of the Skew factor (S_i) on the peak.

in Fig. 11.

$$f(x) = \frac{Amp_i}{S_i} \tan^{-1} \left(\frac{S_i \sin(x)}{1 - S_i \cos(x)} \right), \quad (19)$$

The RSSI values are mapped to (19) using (20).

$$\phi = \left(\frac{\pi}{P_i^+ - P_i^-} \right) P_i^R - P_i^-, \quad (20)$$

Finally, the solution is given in (21), using a piecewise function to isolate the peak and range according to P^+ and P^- .

$$d_i = \begin{cases} \sqrt{\left(\frac{(m_i+1)A_r h_i^{(m_i)}}{2\pi} \right) \frac{P_i^T}{P_i^R}}, & P^+ \leq P_i^R \leq P^-, \\ \sqrt{\left(\frac{(m_i+1)A_r h_i^{(m_i)}}{2\pi} \right) \frac{P_i^T}{P_i^R}} + f(\phi), & P^- \leq P_i^R \leq P^+, \end{cases} \quad (21)$$

B. Estimating the Halo Region

The HR is taken to be the Euclidian distances at which the RSSI values deviate from the Lambertian model. However, this region is not well distinguished by the available data and varies slightly among the light sources, indicating further potential irregularities between lights. This is potentially caused by the varying subcarrier frequencies, environmental factors, or practical dissimilarities such as the simplistic COB source mounting method in Fig. 4d. The author notes that for the given dataset resolution, the HR bounds are subjective, and therefore a quantitative measure for calculating the HR cannot be provided. However, the following procedure is proposed and applied within this study. Algorithm 3 builds on the notion that by ignoring the values within, and slightly around, the HR, the Lambertian model can correctly fit all other data points. Once compared against all measurements, this emphasises the HR where measurements do not adhere to the ideal Lambertian fit. For this analysis, the authors apply the exaggerated estimates $D_{init}^* = [1.285, 1.415]$ as indicated by vertical black lines in Fig. 9.³ Algorithm 3 Step 5 obtains the $\psi_{1/2}^i$ and sub_i values given in the second and third rows of Table VII. Fig. 12 presents the results of step 6 of Algorithm 3 where the UCP_i data is used to plot the Lambertian relationship against the subcarrier specific measurements. The HR in Fig. 12 is amplified and clearer to identify when compared against the similar plots in Fig. 9. Through visual inspection, the final estimate of $D = [1.305, 1.412]$ can be obtained from Fig. 12, as presented by dashed vertical lines. Finally, D can be applied to obtain the corresponding P_i^r of the HR for the respective UCP_i values. These are given in rows four and five of Table VII.

³With the data available, the D_{init}^* estimates bounding the Halo-Region are subjective and therefore visually inspected and estimated by the authors.

Algorithm 3: Halo Region Estimation.

1. **Fit:** Determine initial CP using Algorithm 1 and the given $CDF_i^{x,y}/s$
2. **Plot:** Plot the Lambertian model of CP values against calibration data
3. **Estimate:** Make an initial assumption of HR bounds D_{init} . (HR approximated by the distances at which the RSSI are regularly more significant than the expected Lambertian RSSI)
4. **Buffer region:** Exaggerate the assumed HR bounds to increase the assumed region⁴ D_{init}^* .
5. **Re-fit:** While ignoring all measurements within the exaggerated HR, repeat Algorithm 1 to obtain updated $CP(UCP)$ values.
6. **Re-Plot:** Repeat 2. - Plotting the new Lambertian model against the data highlights the HR and more accurate bounds of the HR can be identified.
7. **Inspect:** Estimate the common HR bounds D^- and D^+ among all lights
8. **Power range:** Using the determined D values, obtain the corresponding P_i^r for each light.

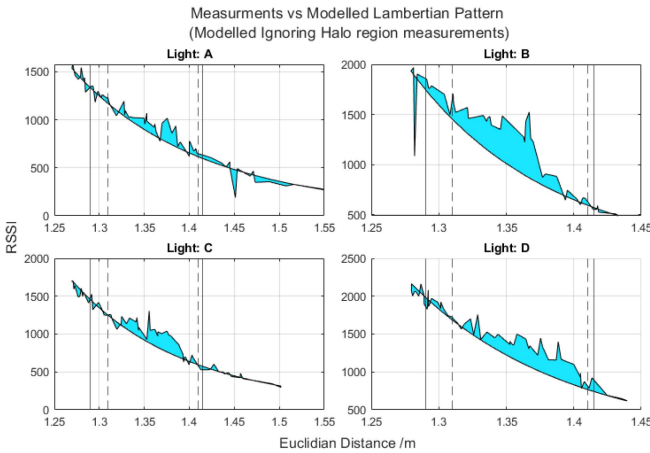


Fig. 12. Optimal Lambertian fits that ignore measurements within the exaggerated HR compared to measurements. Dashed and solid lines represent D and D_{init}^* respectively.

C. Optimal Amplitude and Skew

The CP_i retrieved from step 6 of Algorithm 3 are subsequently used to determine the optimal amp_i and S_i values through the exhaustive search process given in Algorithm 4. The amp_i and S_i results obtained from Algorithm 4 finalise the collection of necessary UCP_i values and can be found in rows six and seven of Table VII.

VII. CALIBRATED RESULTS AND ANALYSIS

Applying the UCP_i values from Table VII, range estimation and multilateration can be performed by replacing (13) with (21). Fig. 13 illustrates the proposed HLC relationship compared to the calibration data. For reference, the original Lambertian model is also compared to the calibration dataset. Notably, the arch in the proposed model accommodates the

TABLE VII
UPDATED CALIBRATION PARAMETERS (UCP)

i	A	B	C	D
$\psi_{1/2}^i$	25.6000	21.0000	23.4000	22.6000
Sub_i	64.0000	12.0000	31.0000	3.000
P_i^r	1206.8200	1522.4400	1295.3300	1757.3300
P_i^+	607.5800	587.4000	585.5000	757.7000
amp_i	0.0236	0.0304	0.0300	0.0300
S_i	0.3831	-0.9204	0.0149	0.3831

Algorithm 4: Calibration: Absolute Search Optimal UCP_i Search.

INPUT: P_i^r , $\psi_{1/2}^i$, sub_i and D for all lights, from Algorithm 3.

OUTPUT: amp_i , S_i for all lights

Determine P_i^T

Calculate m_i using $\psi_{1/2}^i$ in (2).

Determine P_i^{REF} from (12) & (13) using sub_i .

Calculate P_i^T from (11) and (3).

For $amp_i = [0:0.01:0.4]$:

For $S_i = [-0.99:0.001:0.99]$:

Determine Average Error

For each (x, y) in the set of N_i retrieved measurements:

Attain $P_i^{R(x,y)}$ from the average RSSI of $CDF_i^{x,y}(:, sub_i)$.

Use (21) to estimate the distance $\hat{d}_i^{x,y}$.

Determine range error $\varepsilon_{Range,i}^{x,y}$ from (15).

End

Determine the mean range error (16).

End

End

Ascertain the amp_i and S_i values that yield the minimum mean range error

HR better than the Lambertian fit. Comparing Table V and Table VII reveals a decrease in $\psi_{1/2}^i$ values, as expected from ignoring the higher power HR. The subcarrier values generally tend to decrease too. This occurrence may be due to the higher powers of the lower subcarrier frequencies, which are more affected by the physical environment, creating more significant variability in measurements that applies better to a narrower beam angle, indicated by the higher order of Lambertian emission. This can be observed in Fig. 13, where the proposed model fit appears to have more deviation from the measurements. However, this is associated to the greater variation in RSSI at lower subcarrier frequencies.

A. Range Estimation Error

The mean range errors for each light are given in Table VIII using the average RSSI from each $MDF_i^{x,y}(:, sub_i)$. This includes a comparison to the previous Lambertian model results, provided in Table VI. Overall, the proposed HLC solution leads to a consistent reduction in error across all lights with mean range errors of 5.9 mm, 6.0 mm, 6.9 mm, and 5.4 mm, producing mean range error reductions of 40%, 55%,

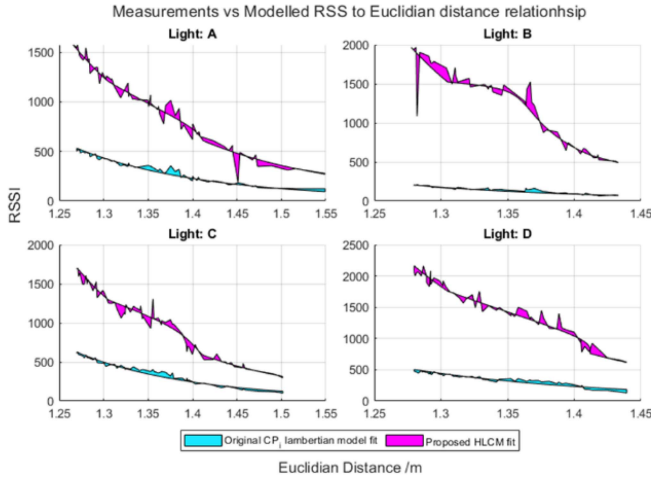


Fig. 13. Original CP_i Lambertian relationship fit vs proposed (HLC) model fit.

TABLE VIII
INDIVIDUAL LIGHT RANGE ERROR RESULTS

A		B		C		D	
Mean /cm	Std. /cm	Mean /cm	Std. /cm	Mean /cm	Std. /cm	Mean /cm	Std. /cm
Lambertian Model - using CP_i							
0.99	1.35	1.33	1.82	1.03	1.36	1.28	1.77
Proposed Model – HLC							
0.59	0.84	0.60	0.81	0.69	0.77	0.54	0.73
Percentage Decrease in Range Error (%)							
40.15	38.19	54.51	55.61	33.10	43.22	57.95	58.77

33% and 58% for lights A, B, C and D respectively. Due to these reductions, light A is no longer the highest performing light system. Notably, in Fig. 14, which illustrates the source range errors, the Halo ring effect is significantly reduced, and range errors become more uniform across all points. The standard deviation of mean range errors for all subcarrier options at the optimal $\psi_{1/2}^i$ value is 2.0 mm, 4.4 mm, 2.6 mm and 4.9 mm for lights A, B, C and D respectively. While this is an increase from those reported in Section V-A, the effects of the subcarrier selection appear minimal.

B. Position Estimation Error

Comparative PE results in Table IX indicate substantial improvements in PE when applying the Proposed Halo lens model. Compared with the previously calibrated Lambertian model, the mean PE for averaged and individual measurement sets achieve a 50% and 39% reduction. Individual measurement sets naturally show a higher error due to the higher noise expected in individual RSSI measurements; however, PE are still less than 5cm. HLC results present a vast improvement on the direct application of datasheet values, reaching mean PE improvements of 59% and 37% for averaged and individual datasets respectively.

Fig. 15 illustrates the comparison of position estimates from both Lambertian and the proposed HLC method.

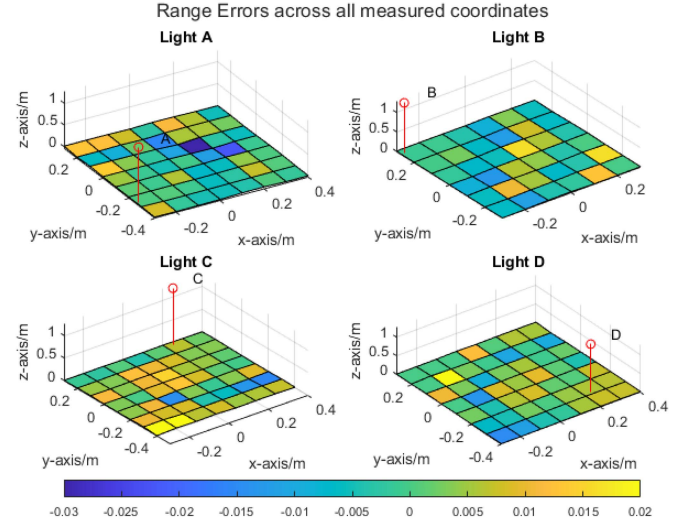


Fig. 14. Range errors for individual lights using the proposed Halo Lens Compensation (HLC) method.

TABLE IX
MULTILATERATION PE RESULTS

Averaged Measurements				Individual Measurements			
Mean /cm	Min /cm	Max /cm	Std. /cm	Mean /cm	Min /cm	Max /cm	Std. /cm
Lambertian Model – using Datasheet $\psi_{1/2}$ (Control)							
7.37	1.10	14.13	3.24	7.37	1.67	12.77	3.15
Lambertian Model – using CP_i							
6.10	0.59	25.77	4.05	7.55	1.10	29.12	5.27
Proposed Model – HLC							
3.06	0.30	8.34	1.78	4.64	0.64	17.79	3.52
HLC vs Control - Percentage Decrease in Range Error (%)							
58.53	72.32	40.95	44.95	37.02	61.74	-39.30	-11.77
HLC vs CP_i - Percentage Decrease in Range Error (%)							
49.88	48.73	67.63	55.99	38.51	41.92	38.91	33.23

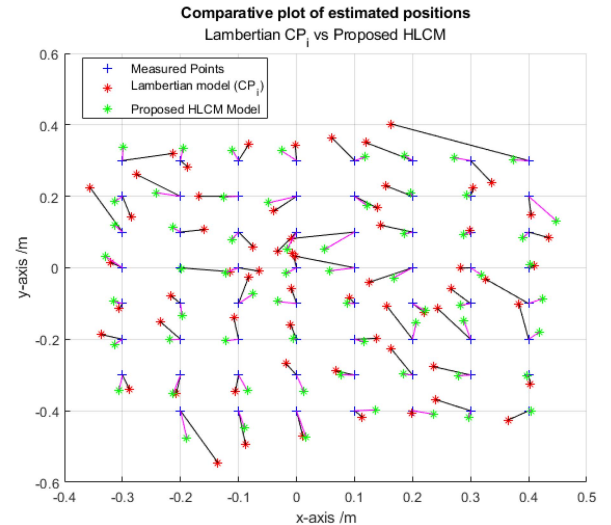


Fig. 15. Plot of measured coordinates and position estimates for both Lambertian and proposed Halo Lens Compensation (HLC) method.

VIII. CONCLUSION

This study underlines the importance of source characteristics, primarily lens structure, on VLP performance. This

emphasises that while Lambertian models may still serve well for simulations, it is crucial to suitably calibrate experimental VLP systems. Real-world data is applied and correlated with lens features to develop a lens compensation solution for TIR lenses that emit a Halo ring effect. The proposed Halo Lens Compensation (HLC) method is proven to correct for the irregularities in the emission pattern and deliver average positioning errors of 3.1 cm and 4.6 cm for average and individually applied measurements, respectively. These correlate to a significant reduction of 50% and 39% or 59% and 37%, respectively, compared to fitting the standard Lambertian model or directly applying the source datasheet parameters.

IX. FURTHER WORK

Following this study, further work may be considered in the following areas:

- 1) Development of a more quantitative method for determining the HR and effects of the estimated range D on VLP performance.
- 2) Evaluation of alternative standard lens types and the subsequent distortion. Perhaps leading to a more generalised calibration procedure for VLP sources
- 3) Further analysis of the subcarrier effects on the VLP performance.

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